

Evaluation

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Assessment

- Accuracy: The predicate capability of the classifier
- Effectiveness:
 - Cost to create the model
 - Cost to use the model
- Robustness: ability to solve noise or missing value
- Scale: Efficiency with Big Data
- Understandable
- Other properties: Tree size, number of laws, quality of law...

Accuracy

- The dataset is divided into two completely independent parts.
 - Training set
 - Test set
- Measures to evaluate accuracy: confusion matrix, fault rate,...
- The method of estimating the accuracy of the classifier:
 - Holdout method, Random Subsampling
 - Cross-validation
 - Bootstrap

Some concepts (1/2)

- Let's:
 - Positive tuples are samples belonging to a major class that is concerned
 - Negative tuples are the models that belong to the remaining classes
- P is the number of positive sample, N is number of negative samples in the test set.
- TP (True Positives): number of positive samples are classified correctly
- TN (True Negatives): number of negative samples are classified correctly

Some concepts (2/2)

- FP (False Positives): number of negative samples are classified incorrectly to positive samples
- FN (False Negatives): number of positive samples are classified incorrectly to negative samples.

		Predicted class		
		+	-	Total
Actual class	+	TP	FN	P
	-	FP	TN	N
Total		P'	N'	P + N

Confusion Matrix

Confusion Matrix

A\P	C	-C	
C	TP	FN	P
-C	FP	TN	N
	P'	N'	All

- Data in computer store, positive samples P are samples with buys_computer = yes

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

- Determine TP, TN, FP, FN?
- Ideally, the sub diagonal should be 0 or approximately 0

Accuracy

- Accuracy: sample rate in test set is classified correctly.

$$accuracy = \frac{TP + TN}{P + N}$$

- Example:

- accuracy = (6954 + 2588) / 10000 = 0.95

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

Error Rate

- Error Rate: The sample rate was incorrectly classified in the test set (= 1 - accuracy)

$$error\ rate = \frac{FP + FN}{P + N}$$

- Example:

- error rate = (412 + 46) / 10000 = 0.05

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

Class imbalance (1/2)

- Classes of interest can be rare compared to other classes
- Example:
 - In the phishing detection application, the class of interest is "fraud" but occurs much less than those of the class "Nonfraudulant".
 - In diagnosis, the class of interest is "cancer", the sample rate is labeled "yes" much lower than the label "no".



Class imbalance (2/2)

- Classifier is correct in negative samples but completely incorrect in positive samples
 - Example:
 - A classifier of accuracy 99% shows a very high probability of prediction. However, if the wrong 1% belongs to the positive sample, 99% becomes pointless
- Resolution by measure sensitivity and specificity

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Sensitivity và Specificity

- Sensitivity: correct positive sample recognition ratio

$$\text{sensitivity} = \frac{TP}{P}$$

- Specificity: correct negative sample recognition ratio

$$\text{specificity} = \frac{TN}{N}$$

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Sensitivity vs. Specificity

- The classifier has a high accuracy of 96.40%.
- However, the ability to identify positive samples is quite low because of low sensitivity.

Classes	yes	no	Total	Recognition (%)
yes	90	210	300	30.00
no	140	9560	9700	98.56
Total	230	9770	10,000	96.40

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Precision vs. Recall

- ◎ Precision: is the proportion of the class that assigns a label to positive is actually positive.

$$precision = \frac{TP}{TP + FP}$$

- ◎ Recall: is the positive sample rate assigned by the classifier.

$$recall = \frac{TP}{TP + FN} = \frac{TP}{P}$$

Precision and Recall

- ◎ Precision(yes) = $90/230 = 39.13\%$
- ◎ Recall(yes) = $90/300 = 30.00\%$

Classes	yes	no	Total
yes	90	210	300
no	140	9560	9700
Total	230	9770	10,000

Precision and Recall

- ◎ Highest precision is 1.0:
 - Showing each sample that the marking class belongs to the positive is actually positive.
 - Unable to show the number of positive samples is classified incorrectly
- ◎ Highest recall is 1.0:
 - Showing all positive samples is labeled properly.
 - Unable to present how many other samples are mislabeled in the positive.

F-Score

- ◎ F-score: A combination of precision and recall

$$F = \frac{2 \times precision \times recall}{precision + recall}$$

- ◎ F-score is harmonic mean between precision and recall
- ◎ Equally weighted between precision and recall ($\beta=1$)
- ◎ If you want to one over another, you can set $\beta=2$, $\beta=0.5$

$$F_{\beta} = \frac{(1 + \beta^2) \times precision \times recall}{\beta^2 \times precision + recall}$$

Example

classes	buy_computer = yes	buy_computer = no	total	recognition(%)
buy_computer = yes	6954	46	7000	99.34
buy_computer = no	412	2588	3000	86.27
total	7366	2634	10000	95.42

F-measure(B-Yes)= 96.81%

Summary of the measurements

AVP	C	~C	
C	TP	FN	P
~C	FP	TN	N
	P'	N'	All

Measure	Formula
accuracy, recognition rate	$\frac{TP + TN}{P + N}$
error rate, misclassification rate	$\frac{FP + FN}{P + N}$
sensitivity, true positive rate, recall	$\frac{TP}{P}$
specificity, true negative rate	$\frac{TN}{N}$
precision	$\frac{TP}{TP + FP}$
$F, F_1, F\text{-score}$, harmonic mean of precision and recall	$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
F_β , where β is a non-negative real number	$\frac{(1 + \beta^2) \times \text{precision} \times \text{recall}}{\beta^2 \times \text{precision} + \text{recall}}$

Exercise 3

- Data relating to the customer classification is deceptive or non-deceptive by a bank before lending:

		Predict		
Actual	Class	Deceptive	Non-deceptive	Total
	Deceptive	44	15	59
	Non-deceptive	20	146	166
	Tổng	64	161	225

- Suppose the class of interest is deceptive, create confusion matrix
- Calculating the measurements accuracy, error rate, sensitivity, specificity, precision, F-Score

Estimation methods



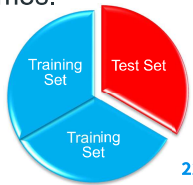
Reliability when estimating

- Whether the figures are calculated from the measurements that are reliable?
 - Depends on the type of data
 - Depend on how the data is collected
 - Depends on how to divide data into training and test episodes.
 - ...

→ Method is required to reliably estimate the accuracy

Holdout Method

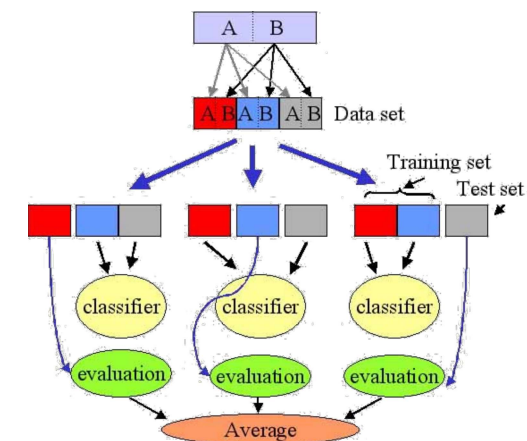
- Data is randomly divided into 2 independent sections
 - Training set up 2/3 to draw the model
 - Test set accounted for 1/3 to estimate accuracy
- Samples may not represent all data, missing class in the experiment set
- Random sampling: is the variant of holdout
 - Repeat holdout k times, the accuracy is the average of all times.



K-fold Cross-Validation

- Randomly divide the data into K-independent and roughly equal size. $D=\{D1, D2, \dots, Dk\}$
 - Perform k evaluation.
 - For the time i, the D_i episode was used as a test, the rest of the training
 - K = 10 commonly used
- Leave-one-out: is k fold with K is the number of samples (only applies when the data size is small)
- Stratified cross-validation: Distributing the classes of samples in each fold is roughly the same as the original data

K-fold Cross-Validation



Bootstrap

- ⊙ Usually apply to small datasets
- ⊙ Each time a sample is selected, it is likely to be picked again and added to the training set
- ⊙ There are a few bootstrap methods, which are common **.632 Bootstrap**
 - The d-size dataset will be sampled Bootstrap d times. So training set has d samples. Samples that do not include the training will be used to test. About 63.2% of data fall into training assignments and 36.8% for test episodes (because according to probability $(1-1/d)^d \approx e^{-1} = 0.368$)
 - Repeats the sampling k times and the accuracy:

$$Acc(M) = \frac{1}{k} \sum_{i=1}^k (0.632 \times Acc(M_i)_{test_set} + 0.368 \times Acc(M_i)_{train_set})$$

The End